Speed, Distance & Time

Year 5 Accelerated — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

Easy: A car travels 240 km in 3 hours. What is its average speed?
Method & answer: Speed = distance ÷ time = 240 ÷ 3 = 80 km/h.

Standard: A train travels at 96 km/h for 2 hours 15 minutes. How far does it travel?
Method & answer: 2 hours 15 minutes = 2.25 hours. Distance = speed × time = 96 × 2.25 = 216 km.

Advanced: A plane covers 1,800 km in 2 hours 30 minutes. What is its average speed?
Method & answer: 2 hours 30 minutes = 2.5 hours. Speed = distance ÷ time = 1,800 ÷ 2.5 = 720 km/h.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A car travels 350 km in 5 hours. What is its speed?

2. A train travels at 125 km/h for 4 hours. How far does it travel? A) 450 km B) 500 km C) 550 km D) 600 km

3. A plane covers 2,100 km in 3 hours. What is its speed?

4. A car travels at 70 mph for 3 hours. How far does it travel?

5. Which formula rearranges D = S × T to find time? A) T = D × S B) T = D ÷ S C) T = S ÷ D D) T = S − D

6. A cyclist rides 84 km in 3.5 hours. What is her speed?

7. A ferry travels at 32 km/h for 2 hours 15 minutes. How far does it travel?

8. A car covers 372 km in 6 hours. What is its speed?

9. Select ALL the speed calculations below that are correct: A) 504 km in 7 hours = 72 km/h B) 315 km in 4.5 hours = 65 km/h C) 760 km in 8 hours = 95 km/h D) 186 km in 3 hours = 62 km/h E) 450 km in 5 hours = 99 km/h

10. A train travels 288 km in 3 hours 12 minutes. What is its speed?

11. A runner covers 26.4 km in 2.2 hours. What is her speed?

12. Which speed would complete a 210 km journey in exactly 3 hours? A) 60 km/h B) 65 km/h C) 70 km/h D) 75 km/h

13. A coach travels at 78 km/h for 2.5 hours. How far does it travel?

14. A car takes 4.25 hours to travel 340 km. What is its average speed?

15. A cyclist travels 27 km in the first 1.5 hours and 33 km in the next 1.5 hours. What is her average speed over the whole journey?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A ship sails 585 km in 7.5 hours. What is its speed?

6. A cyclist travels at 27 km/h for 100 minutes. How far does she travel?

11. A train travels at 144 km/h. Select ALL statements that are TRUE: A) In 2.5 hours it covers 360 km B) In 45 minutes it covers 108 km C) In 20 minutes it covers 50 km D) In 1 hour 15 minutes it covers 180 km E) In 10 minutes it covers 24 km

2. A car travels at 92 km/h for 2 hours 45 minutes. How far does it travel?

3. A train travels 396 km in 4.4 hours. What is its speed? A) 85 km/h B) 88 km/h C) 90 km/h D) 92 km/h

4. A car travels for 40 minutes at 96 km/h. How far does it travel?

5. A plane flies 3,250 km in 5 hours. What is its speed?

7. A lorry travels at 84 km/h for 3.5 hours. How far does it travel? A) 252 km B) 264 km C) 276 km D) 294 km

8. A car covers 425 km in 5 hours. What is its speed?

9. A train travels for 18 minutes at 150 km/h. How far does it travel?

10. A walker travels at 6.5 km/h for 2 hours 24 minutes. How far does he walk?

12. A ferry takes 4.8 hours to travel 336 km. What is its average speed?

13. A coach travels at 66 km/h for 3 hours 10 minutes. How far does it travel?

14. Which journey has the greatest speed? A) 308 km in 4 hours B) 459 km in 5.4 hours C) 372 km in 4 hours D) 440 km in 5 hours

15. A car travels 168 km in the first 2.4 hours and 232 km in the next 2.6 hours. What is its average speed for the whole journey?

Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A car travels 100 km in the first 2 hours and 200 km in the next 3 hours. What is its average speed for the whole journey?

2. A train travels 546 km at an average speed of 78 km/h. How long does the journey take? A) 6 hours B) 6.5 hours C) 7 hours D) 7.5 hours

3. A cyclist travels 36 km in 2 hours, then a further 27 km in 1.5 hours. What is her average speed for the whole journey?

4. A plane flies 2,760 km at a speed of 690 km/h. How long does the flight take?

5. A car travels for 2.5 hours at 84

6. A coach's journey is 336 km and takes 4 hours. Later it returns the same distance but takes 48 minutes longer. What is its speed on the return journey? A) 60 km/h B) 65 km/h C) 70 km/h D) 75 km/h

7. A ship sails at 42 km/h for 3.5 hours, then speeds up to 56 km/h for 2.5 hours. How far does it sail in total?

8. Two cars leave the same place at the same time, travelling in opposite directions. One travels at 78 km/h, the other at 92 km/h. How far apart are they after 3.5 hours?

9. A journey of 585 km is completed at an average speed of 65 km/h. Select ALL of the following that are TRUE: A) The journey takes 9 hours B) In 4 hours, 260 km is covered C) Doubling the speed to 130 km/h would halve the time to 4.5 hours D) At 65 km/h, half the journey (292.5 km) takes 4.5 hours E) In 6 hours, 375 km is covered

10. A car's journey is 630 km. For

11. A cyclist travels 72 km at 24 km/h, rests for 45 minutes, then travels a further 65 km at 26 km/h. How long does the whole trip take, including the rest?

12. A train leaves at 08:50 travelling at 96 km/h. How far has it travelled by 11:20? A) 216 km B) 228 km C) 240 km D) 252 km

13. A car travels 405 km in 4.5 hours for the first part of a journey, then a further 240 km in 3 hours. What is its average speed for the whole journey?

14. A cyclist and a runner both set off from the same point at the same time, travelling in the same direction. The cyclist rides at 32 km/h and the runner jogs at 11 km/h. How far apart are they after 3.5 hours?

15. A lorry travels 720 km. It covers

km/h, then for 1.5 hours at 96 km/h. How far does it travel in total?

the first half it travels at 90 km/h and for the second half at 84 km/h. How long does the whole journey take, to the nearest tenth of an hour?

the first 450 km at 90 km/h. At what speed must it travel the remaining 270 km to complete the whole journey in exactly 9 hours?